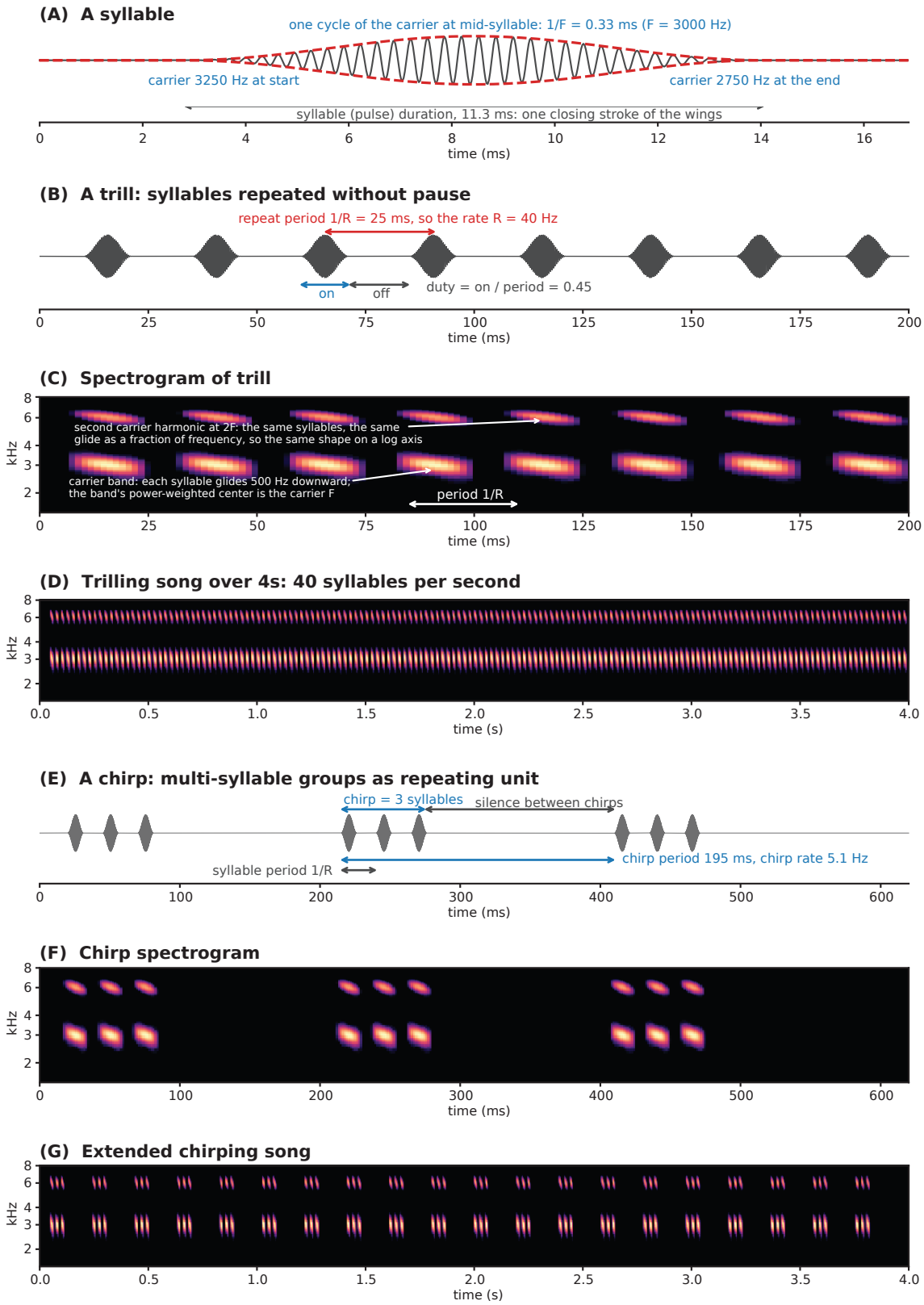

Figure 1. Anatomy of orthopteran songs

Basic features of typical orthopteran songs including continuous trills with regularly repeating syllables in extended bouts and chirps with short bursts of repeated syllables are themselves repeated at a regular interval.

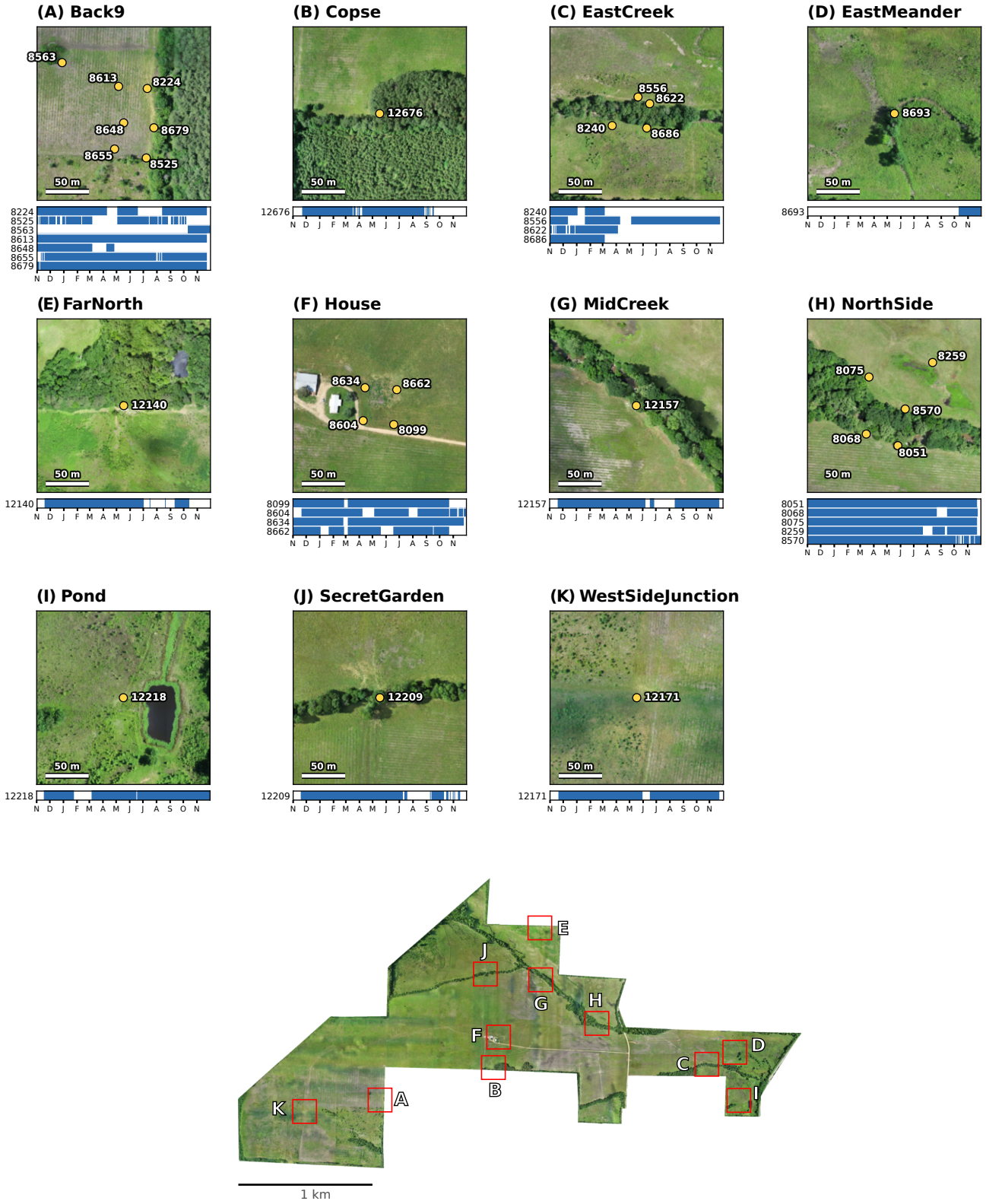

Figure 2. Recorder locations and uptime

All recordings analyzed in this paper were made on Solar BAR (Frontier Labs) autonomous recording units distributed around a 400ha field site in the piney woods and rolling agricultural landscapes of the West Gulf Coastal Plain, in the northern part of the Ark-La-Tex region of the American South.

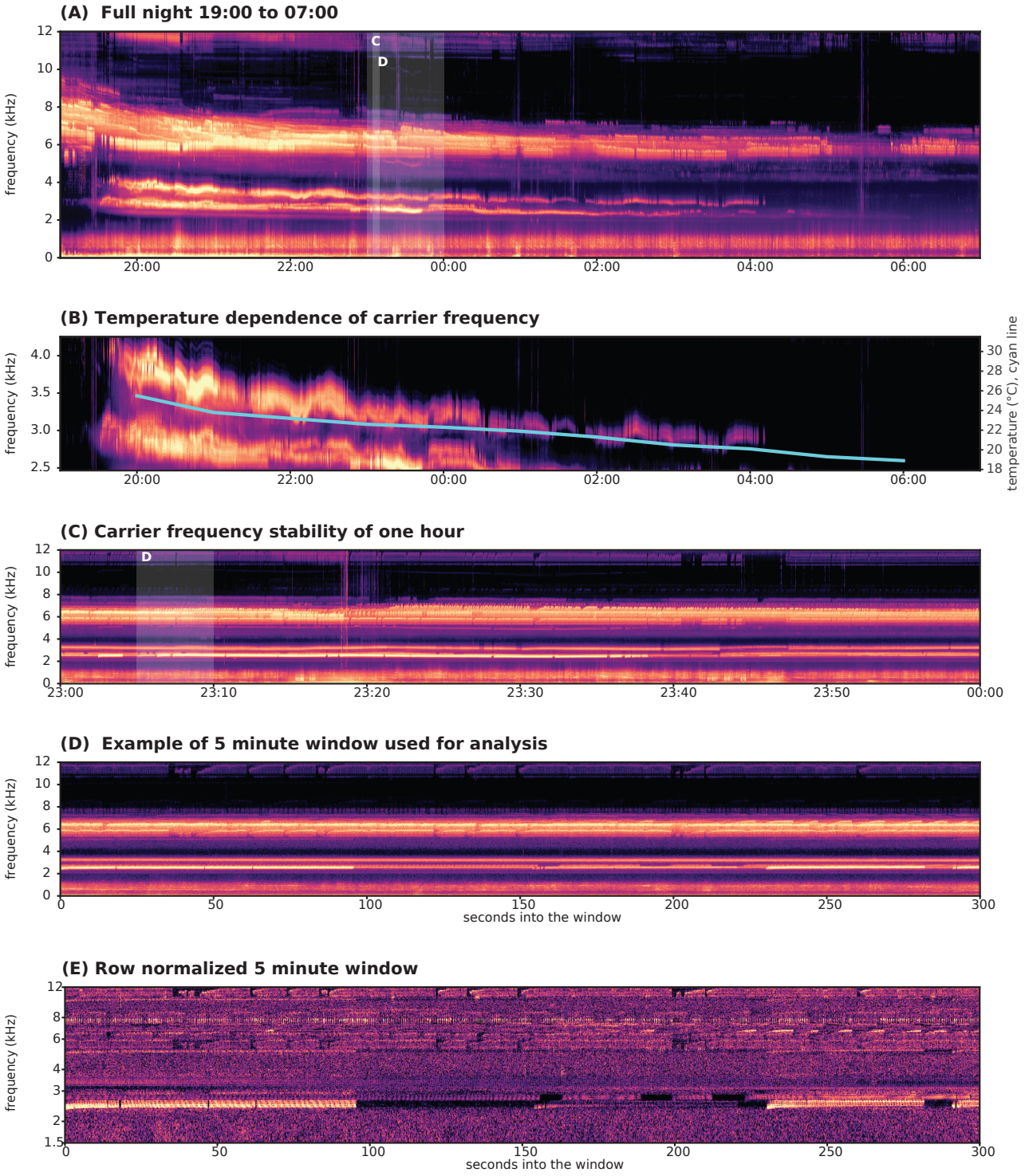

Figure 3. Example night audio.

13 hourlong audio files from recorder BARLT_00008662 on the night of September 20-21, 2025 were combined to cover the period of 7:00 PM to 7:00 AM local (UTC -6) time.

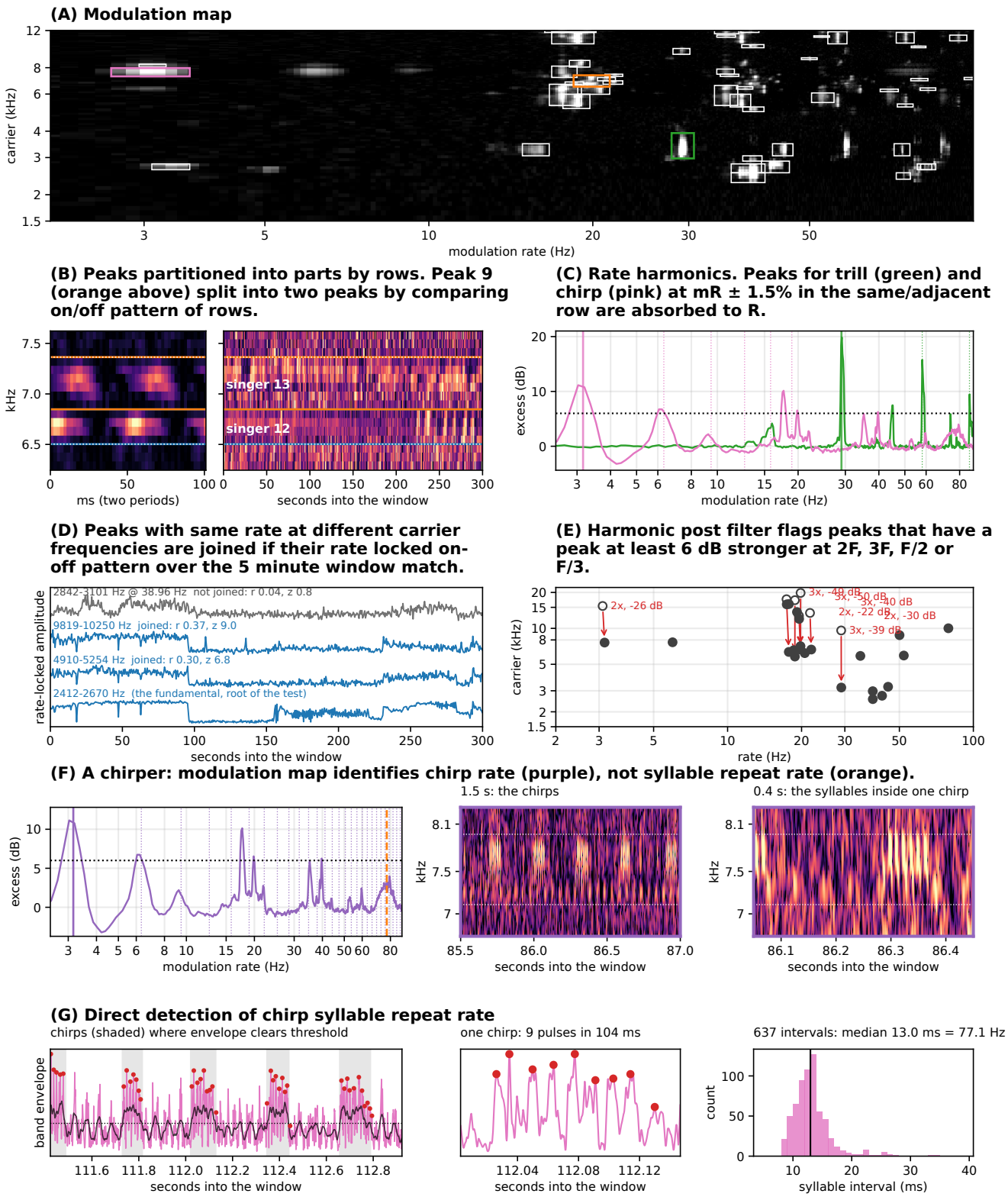

Figure 4. Overview of singer identification process

Data shown are from five minute window starting at 23:05 local time (UTC -6) on September 20, 2025 on the House recorder BARLT_00008662.

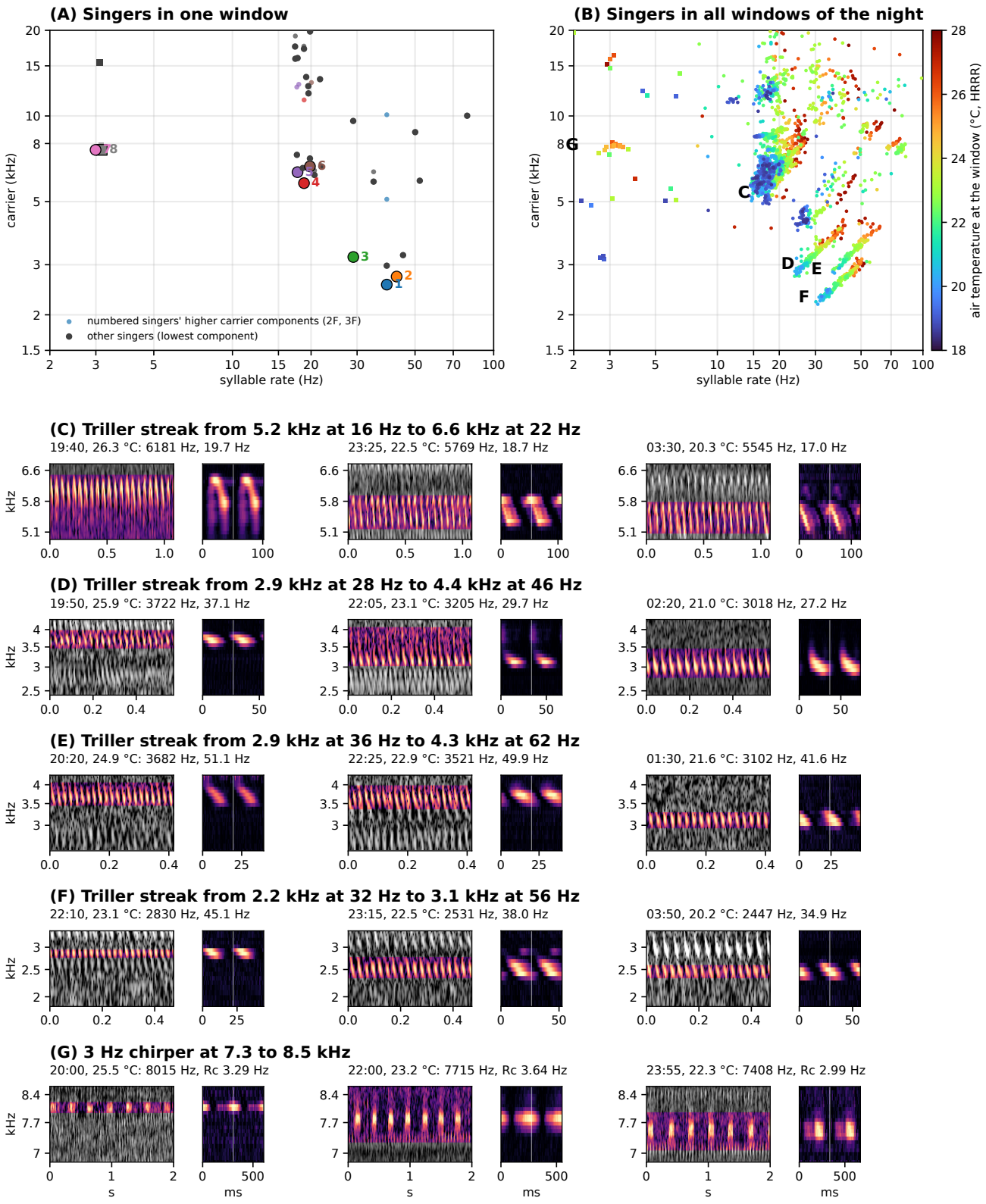

Figure 5. Coherent organization of individual songs across one night

(A) Individual singers identified in window from five minute window starting at 23:05 local time (UTC -6) on September 20, 2025 on the House recorder BARLT_00008662. (B) Singers across night plotted with temperature. (C) to (G) show individual songs with their folded syllable.

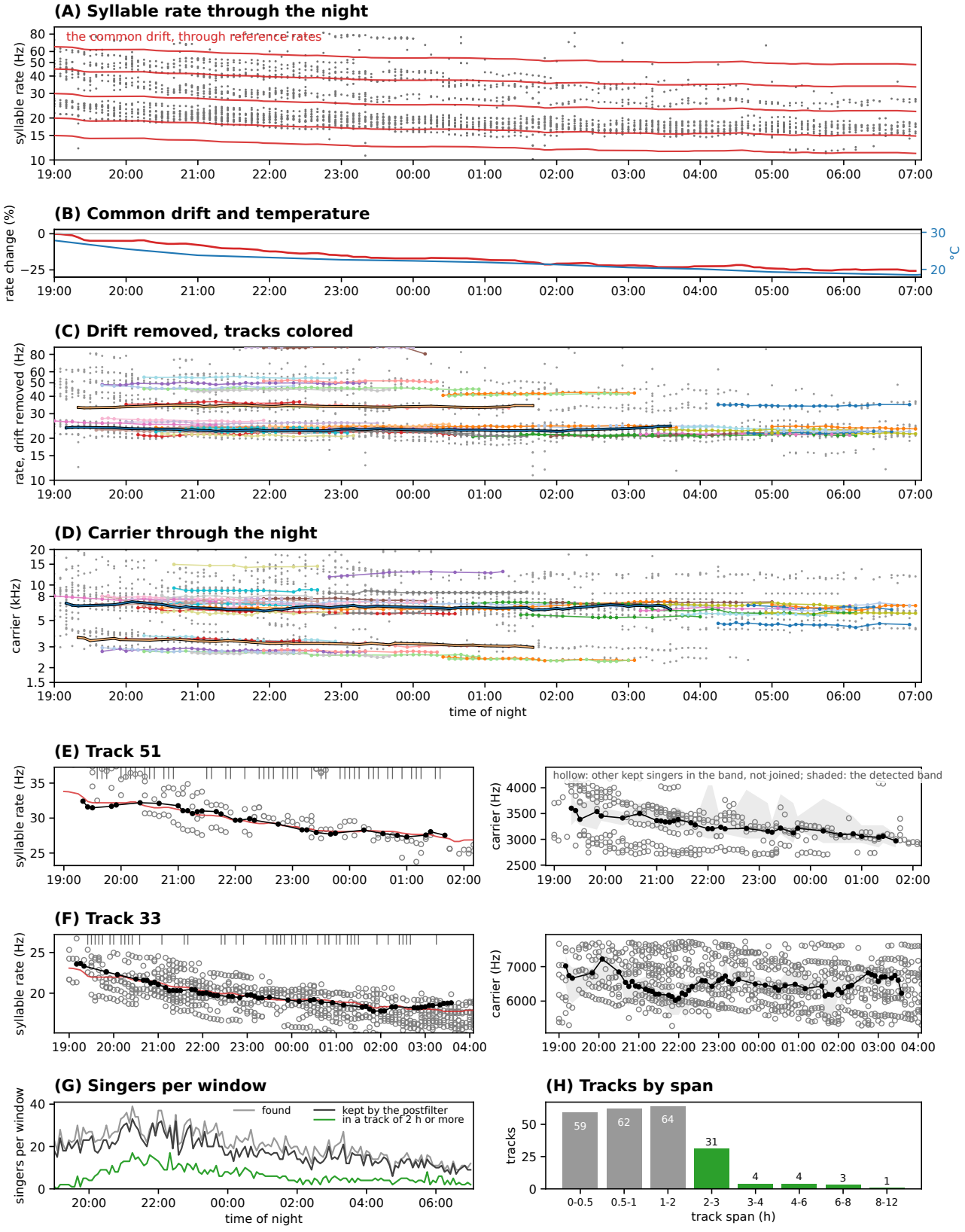

Figure 6. Linking singers across one night

Overview of the process of linking singers including removal of common drift (A,B), connecting tracks by rate (C), and tracking carrier rates (D). Application to individual tracks (E,F), and results (G,H).

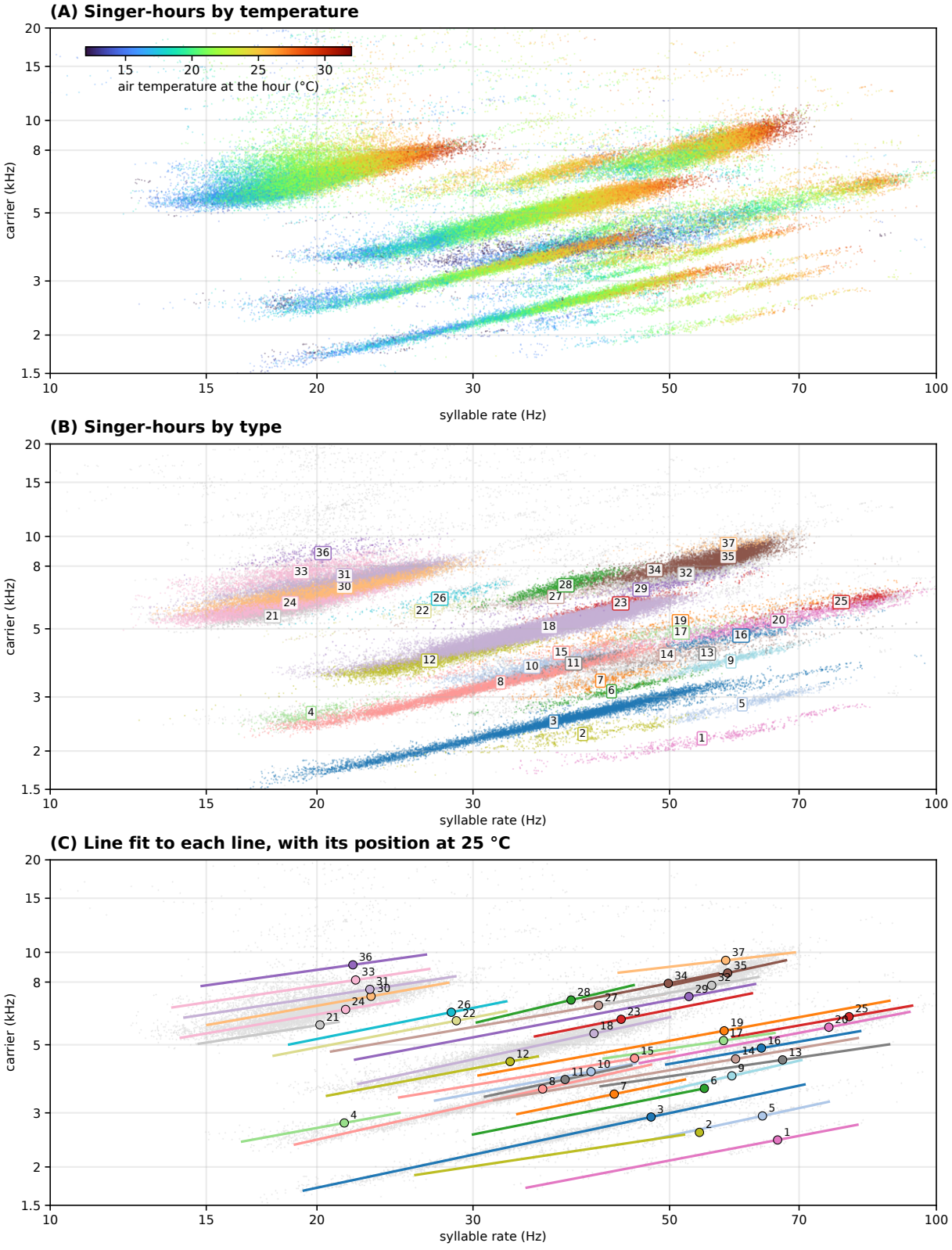

Figure 7. A year of trillers

(A) Point for every triller singer-hour on carrier frequency by syllable-rate plane colored by HRRR 2m air temperature. (B) Classied triller plane types. (C) Lines fit to singers in each type.

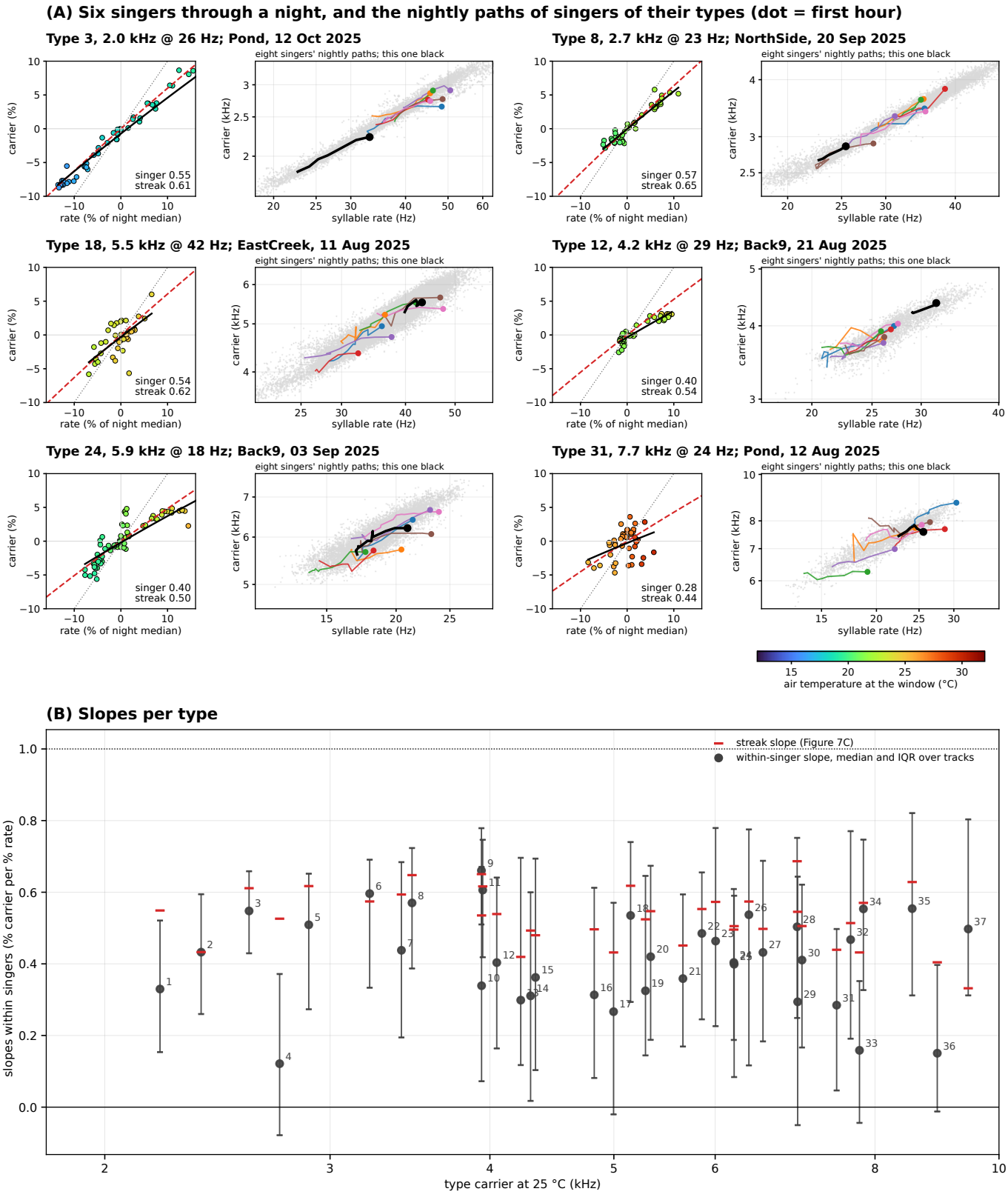

Figure 8. Carrier follows rate within a singer

(A) Individual traced singers broadly follow type line on single nights, and have within night slopes similar to, but slightly lower than type. (B) Slopes of all singers within night by type.

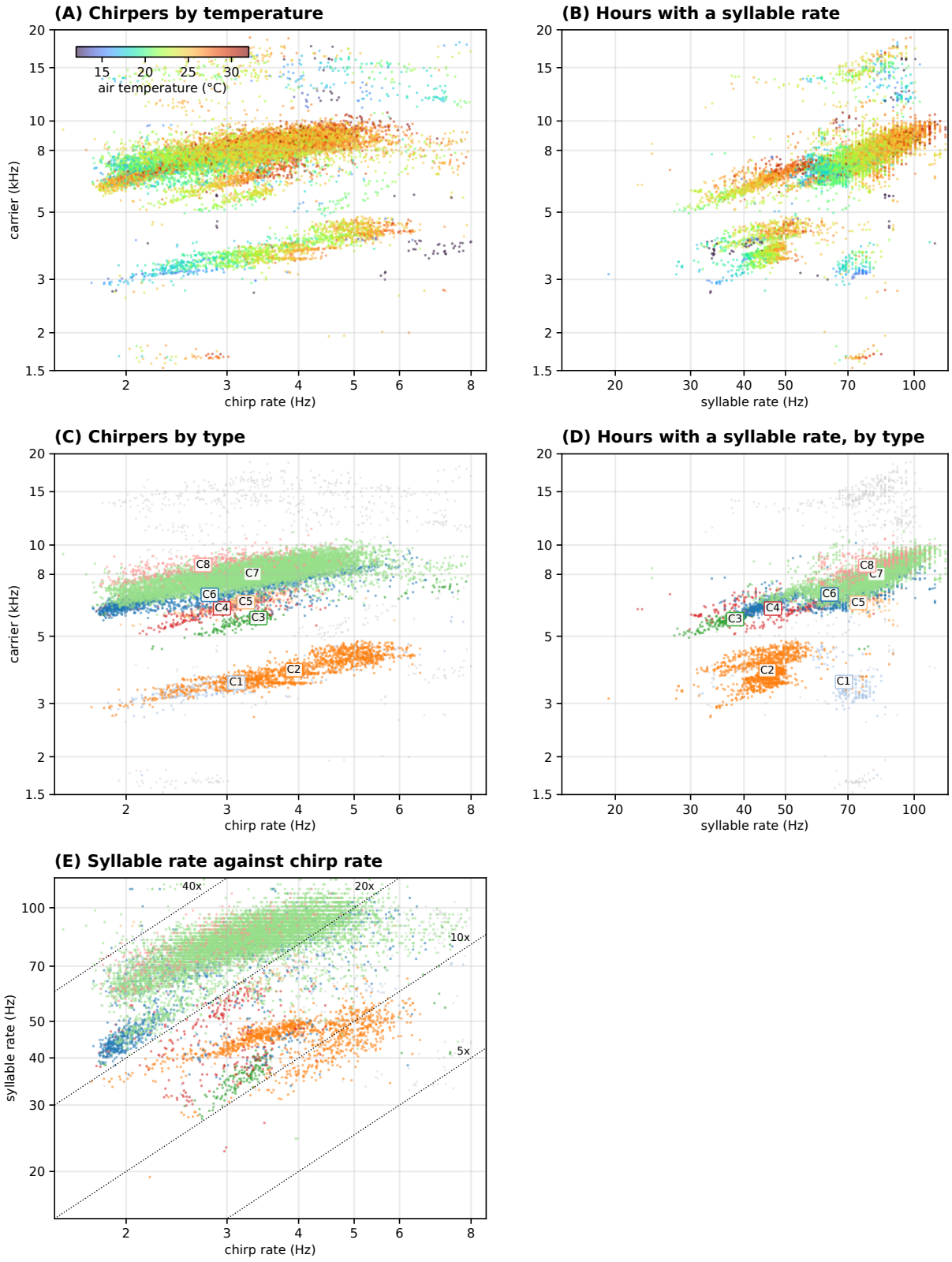

Figure 9. Identification of chirper types

Chirpers have three measured rates: carrier frequency (F), syllable repeat rate (R) and chirp rate (C) and are plotted FxC (A) and FxR planes (B) to facilitate type identification (C-E).

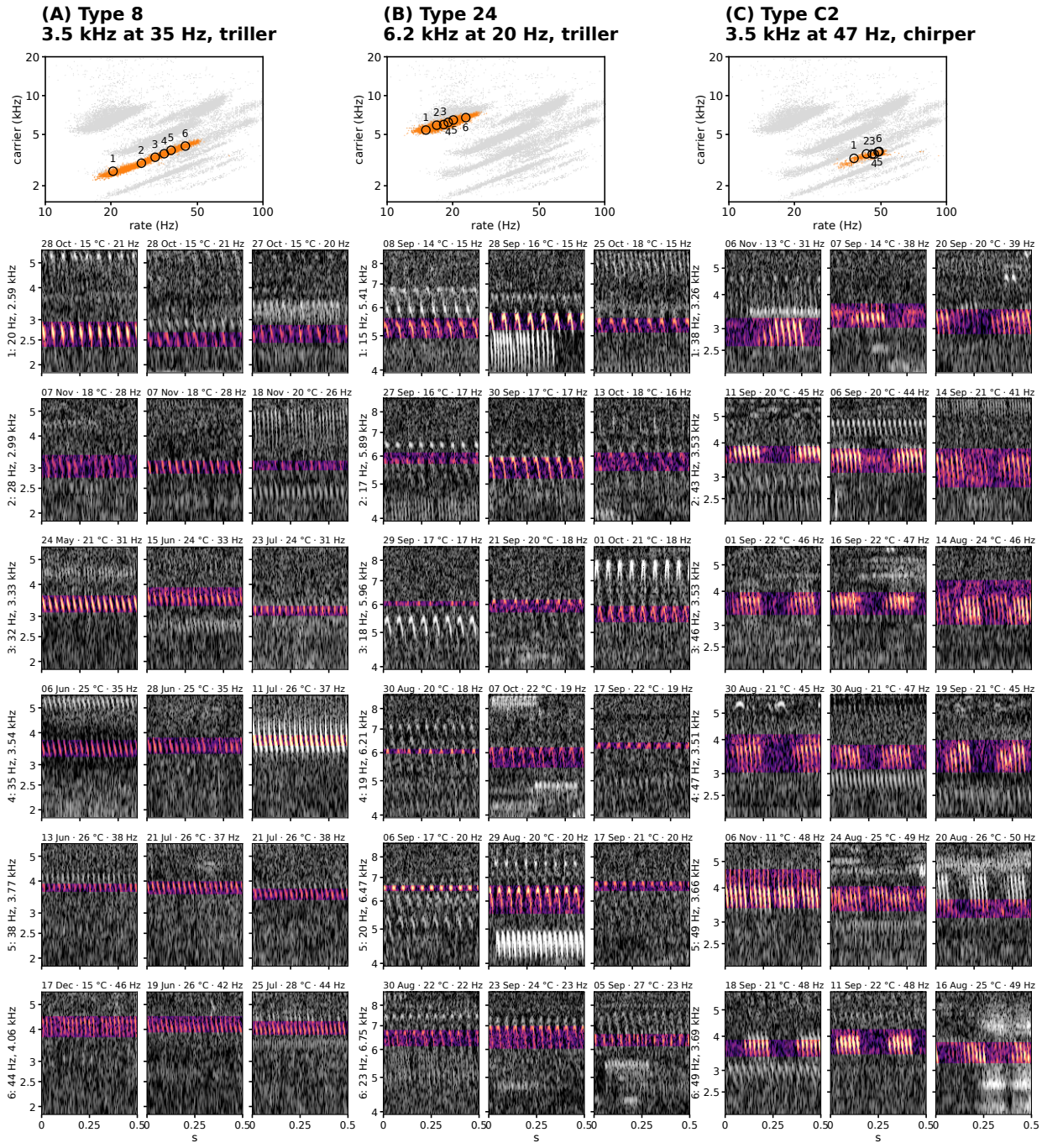

Figure 10. Coherence of three streaks across the year

Three differėnt streaks show coherent song types across temperature range. See also Figure 7S.

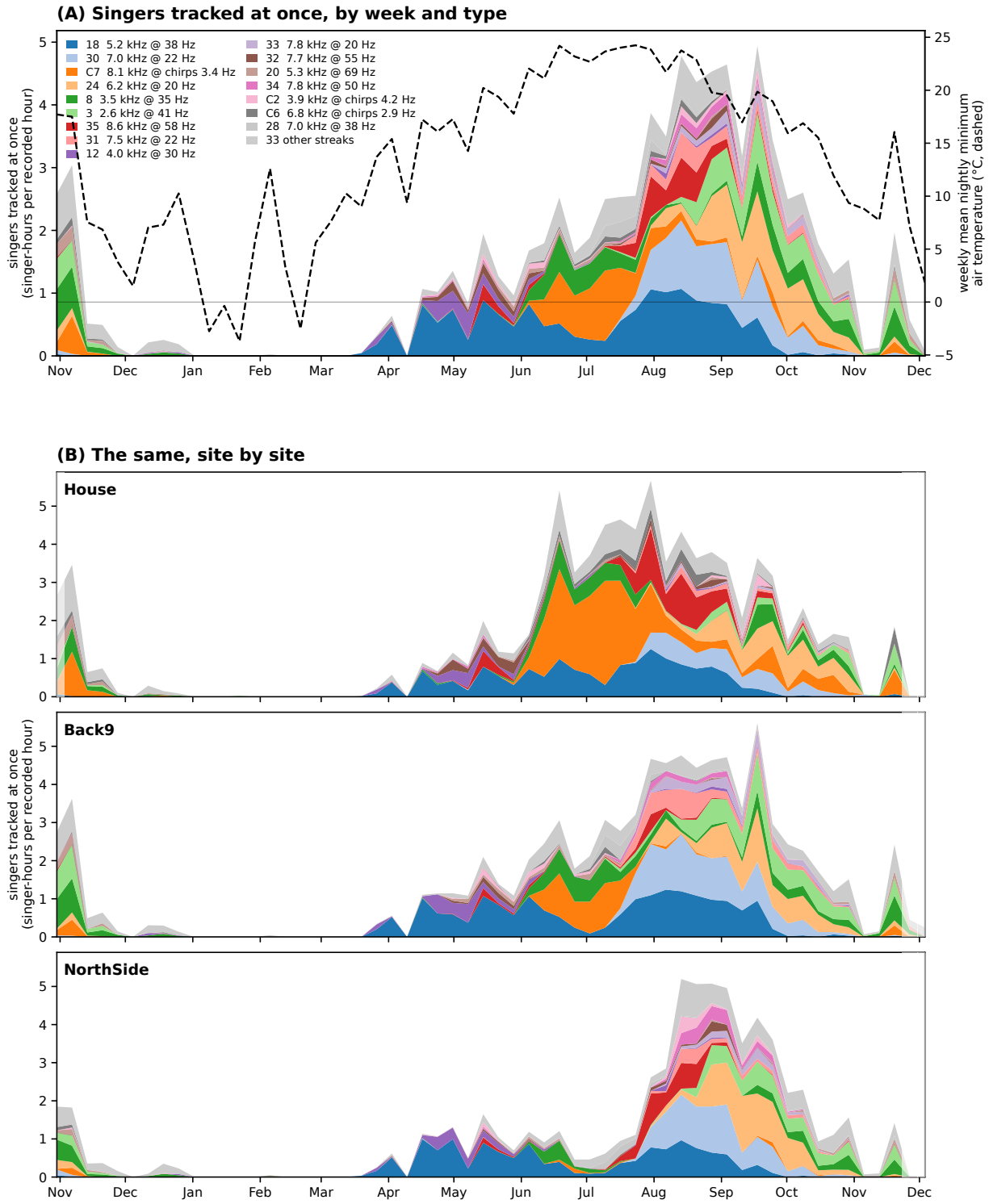

Figure 11. The mix of call types through the year and across sites

Mix of different triller and chirper types across the year (A) and across three sites with multiple recorders (B). The sites are House, located in an open field, Back9, adjacent to a hardwood border to a loblolly pine plantation, and NorthSide in a field alongside tree-lined creek, and adjacent to season pond.

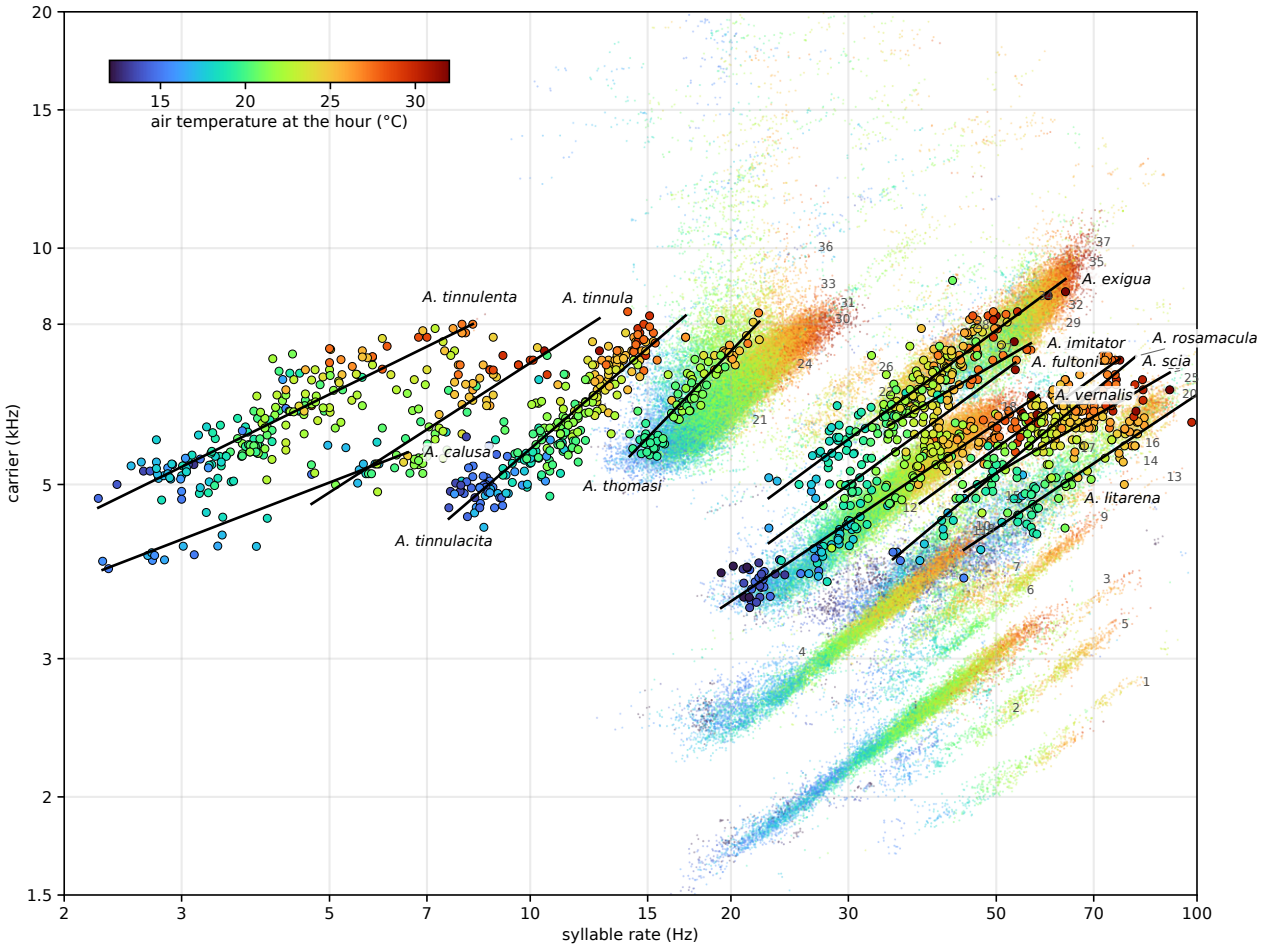

Figure 12. Published *Anaxipha* recordings on the year's plane

The year's trill singer-hours colored by air temperature, as in Figure 7A, with 1,318 recordings of 13 *Anaxipha* species from Walker and Funk (2014; their supplementary song table) as rimmed circles colored by the recording's own temperature on the same scale, each species with its fitted line and name; gray numbers mark the warm end of each type's line.

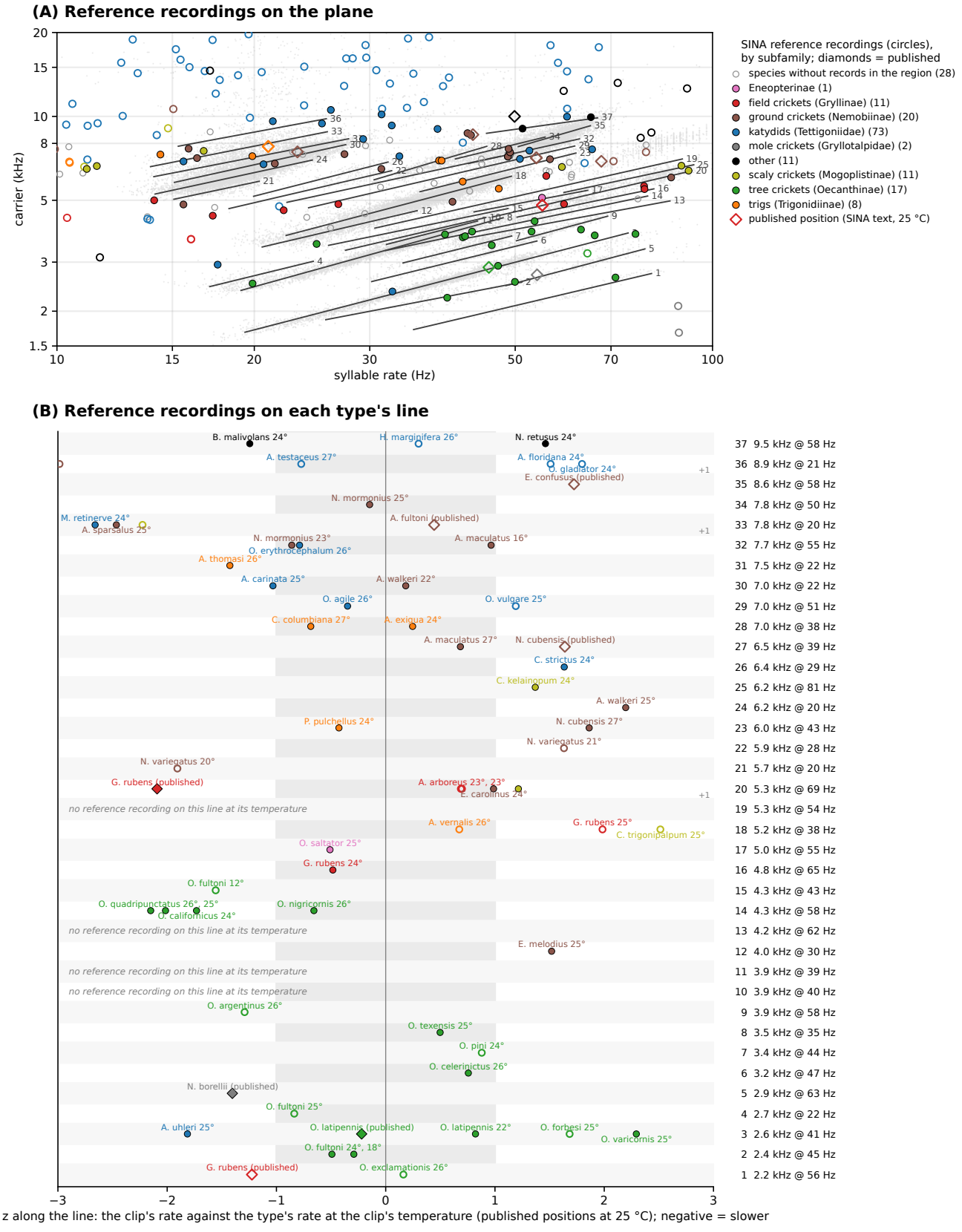

Figure 13. Reference recordings and published positions on the lines

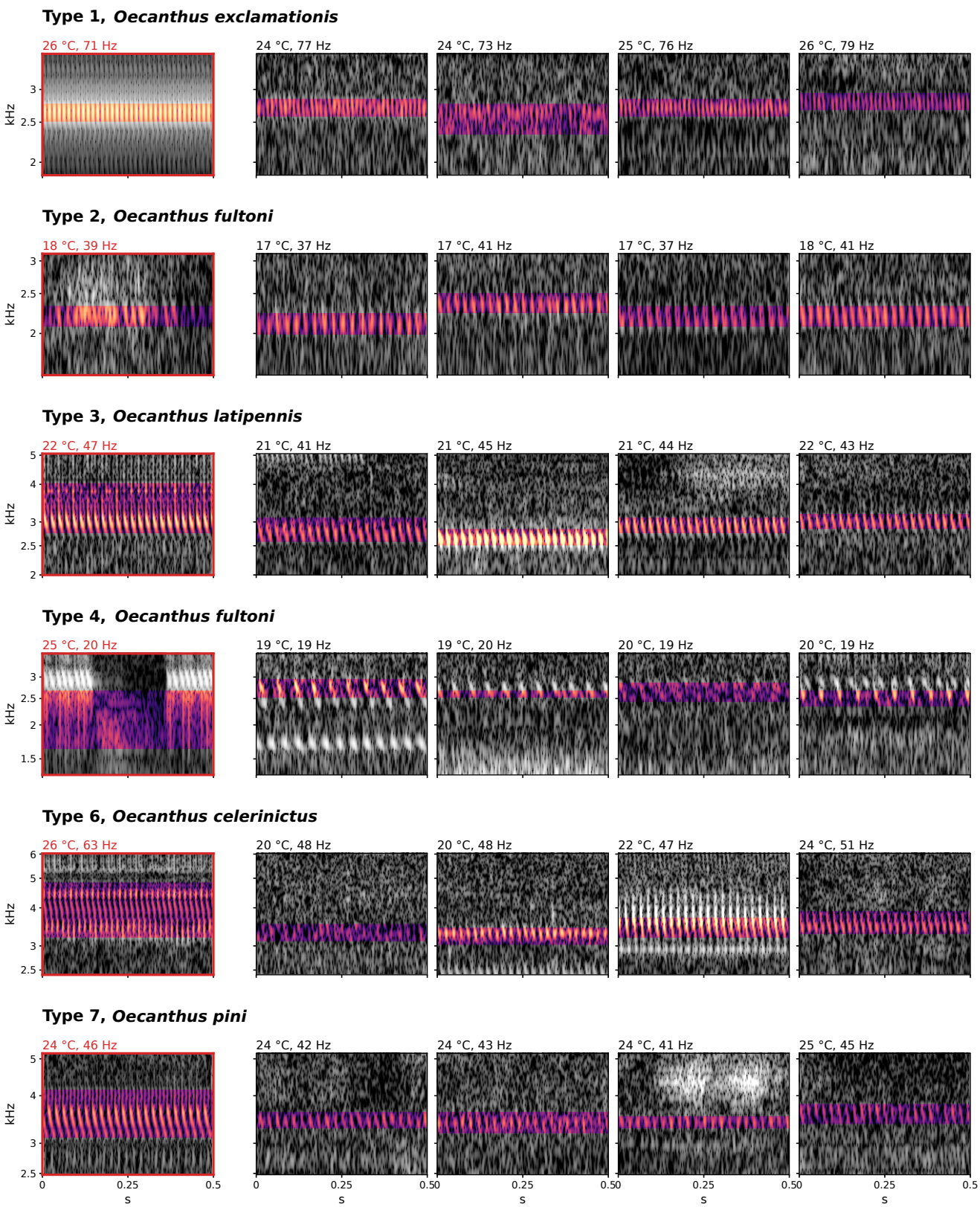

Figure 14.1. Reference recordings beside field singers of the type they match (1 of 7)

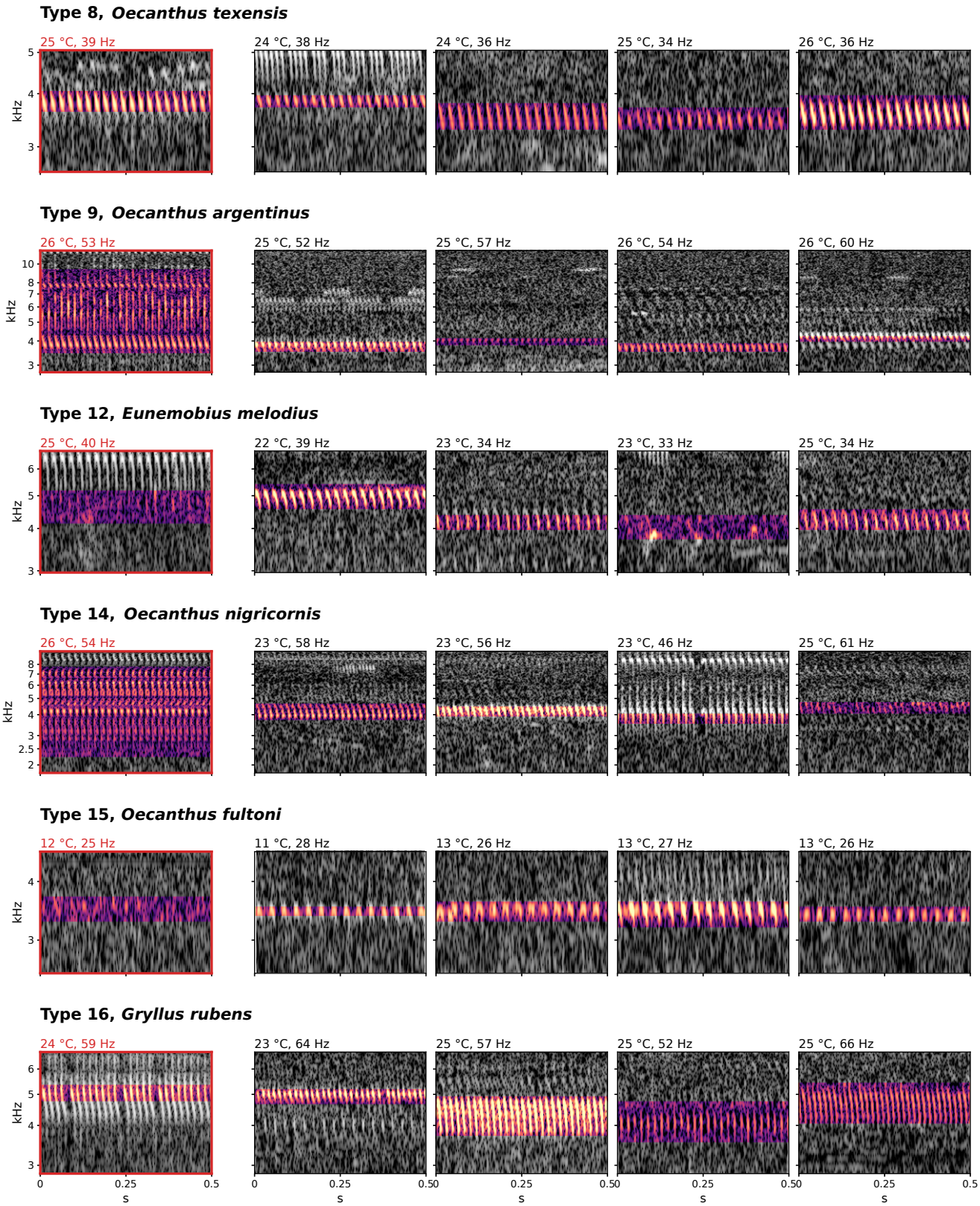

Figure 14.2. Reference recordings beside field singers of the type they match (2 of 7)

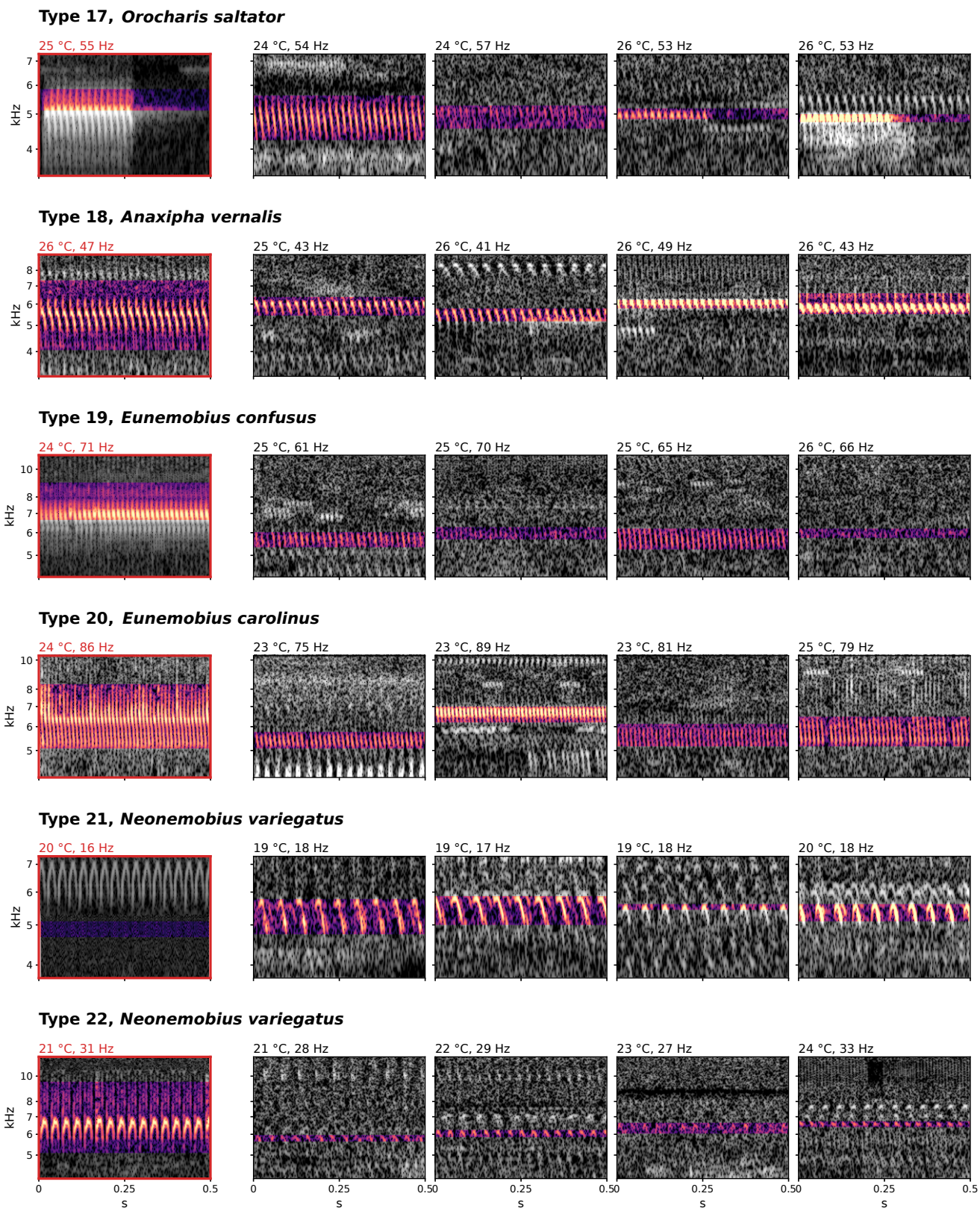

Figure 14.3. Reference recordings beside field singers of the type they match (3 of 7)

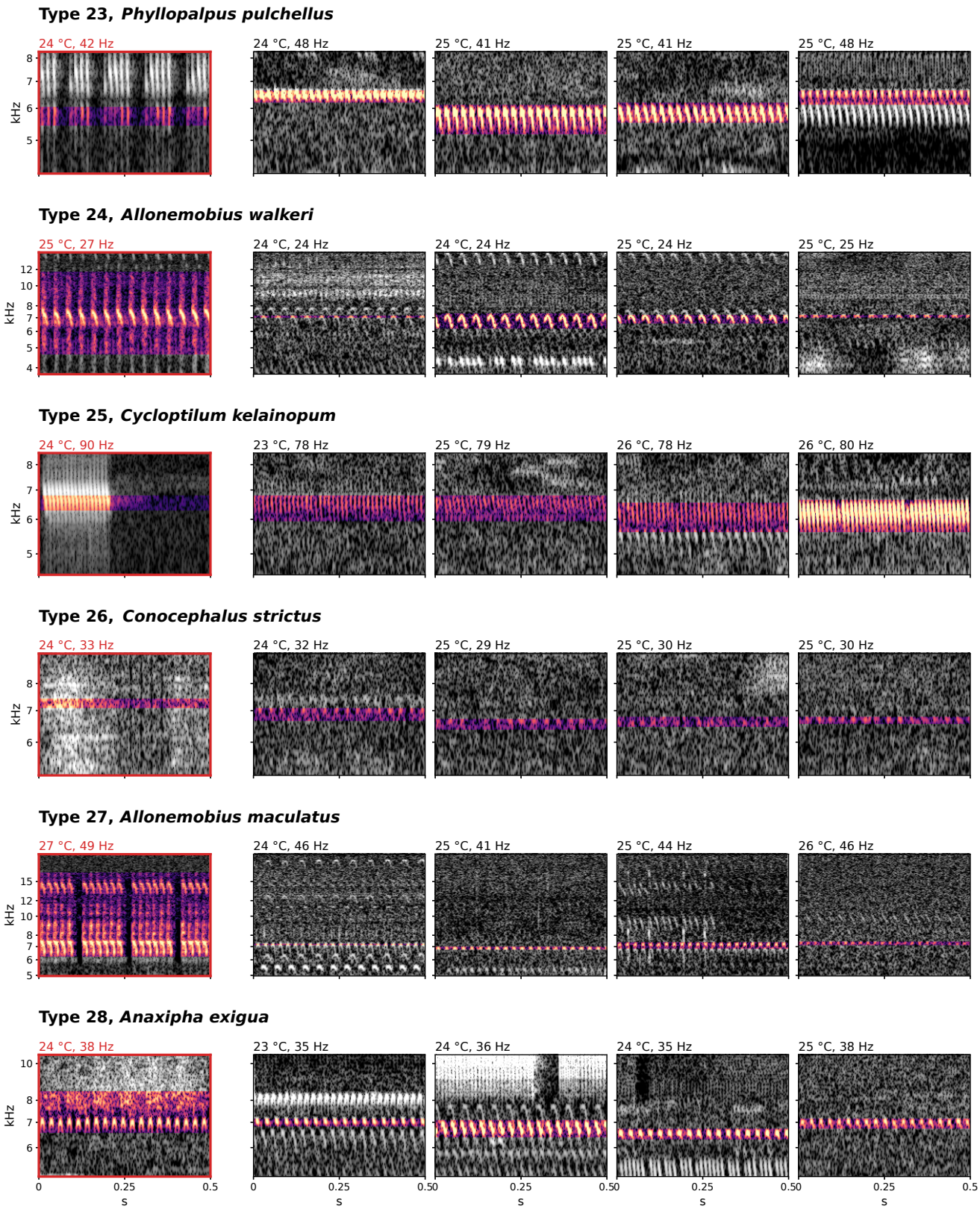

Figure 14.4. Reference recordings beside field singers of the type they match (4 of 7)

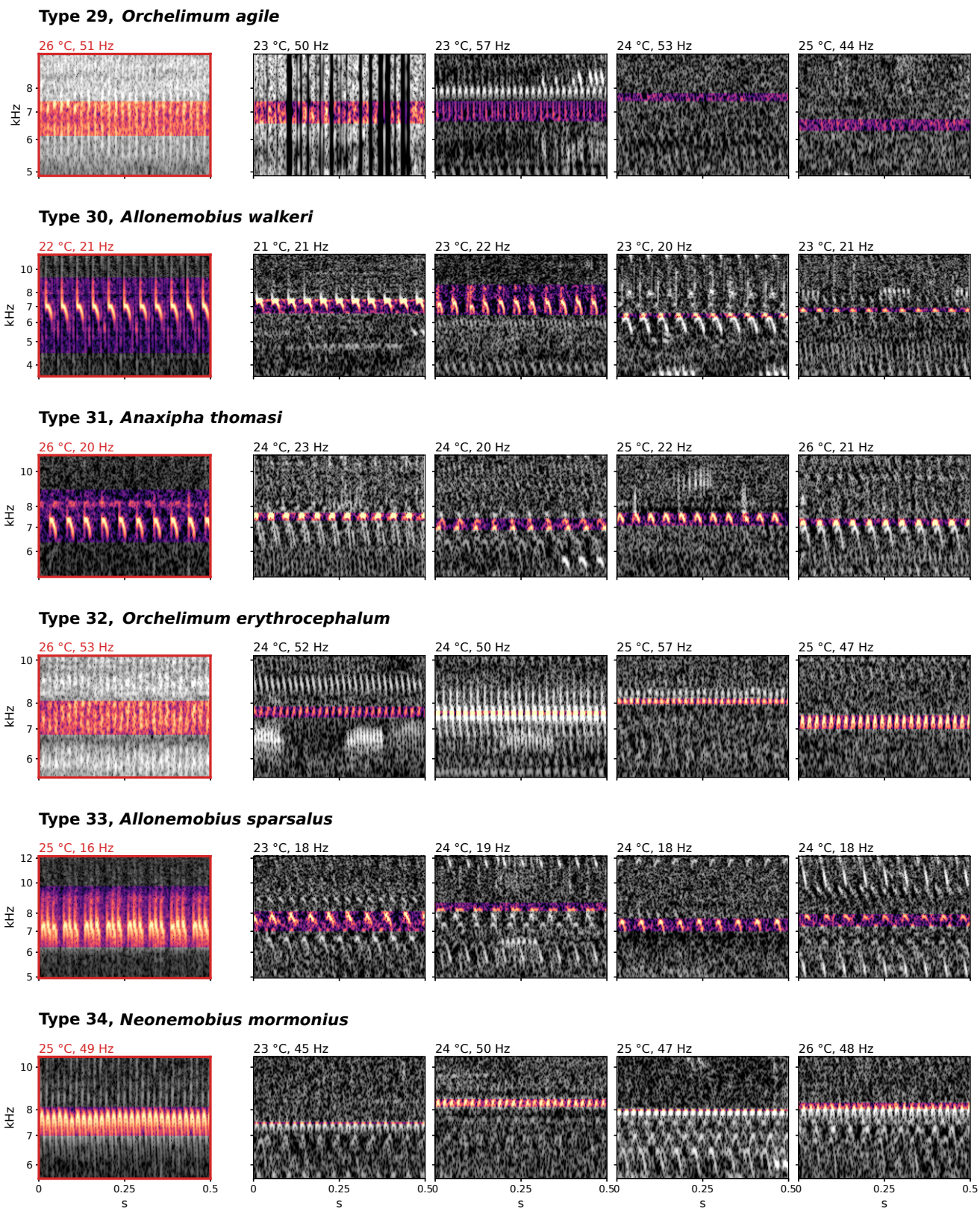

Figure 14.5. Reference recordings beside field singers of the type they match (5 of 7)

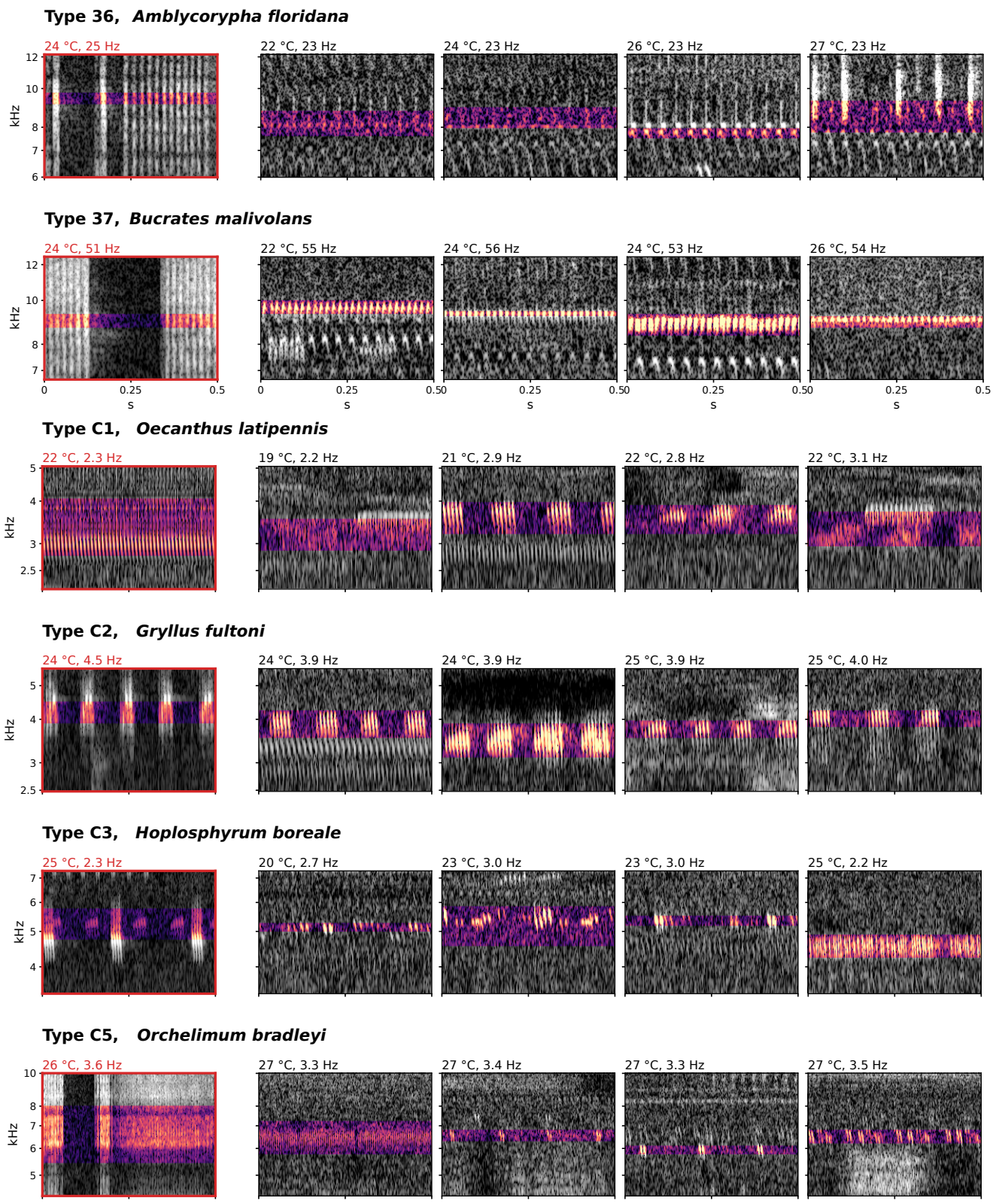

Figure 14.6. Reference recordings beside field singers of the type they match (6 of 7)

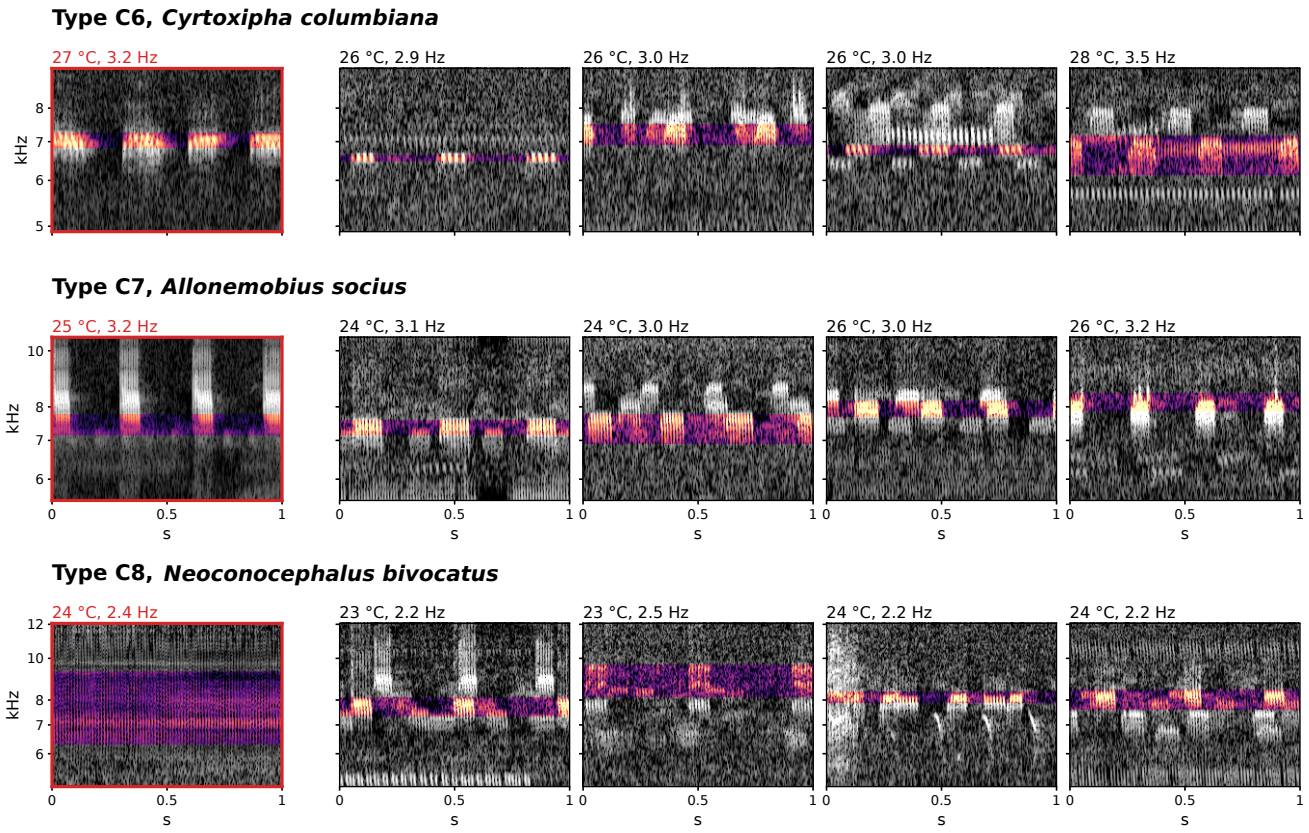

Figure 14.7. Reference recordings beside field singers of the type they match (7 of 7)

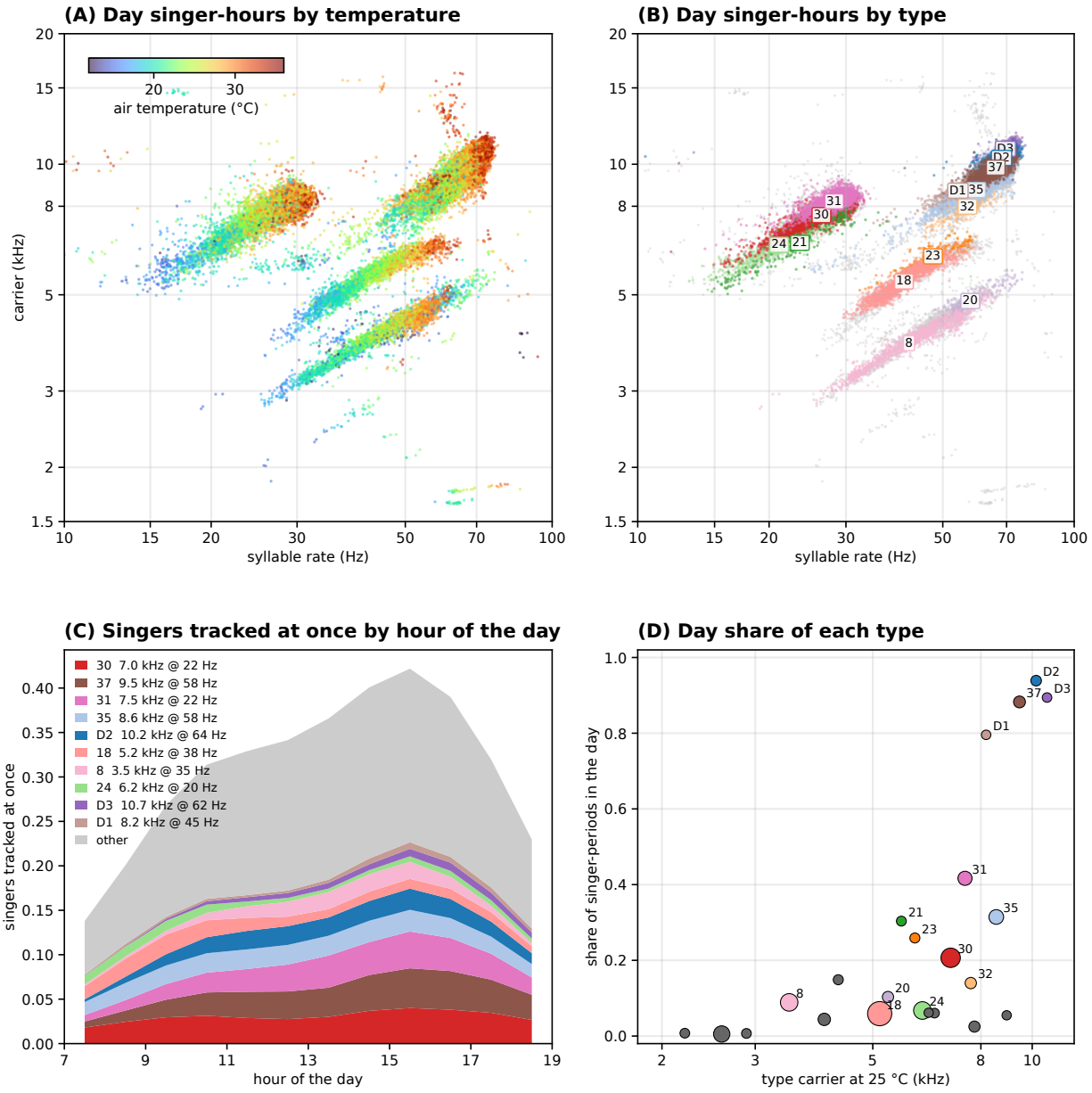

Figure 15. The day singers

(A) Day trillers on carrier frequency x syllable rate plane. (B) Most overlap tracks from night. New tracks labeled with “D”. (C) Daily activity levels of day singers. (D) Fraction of songs of type in day vs night.

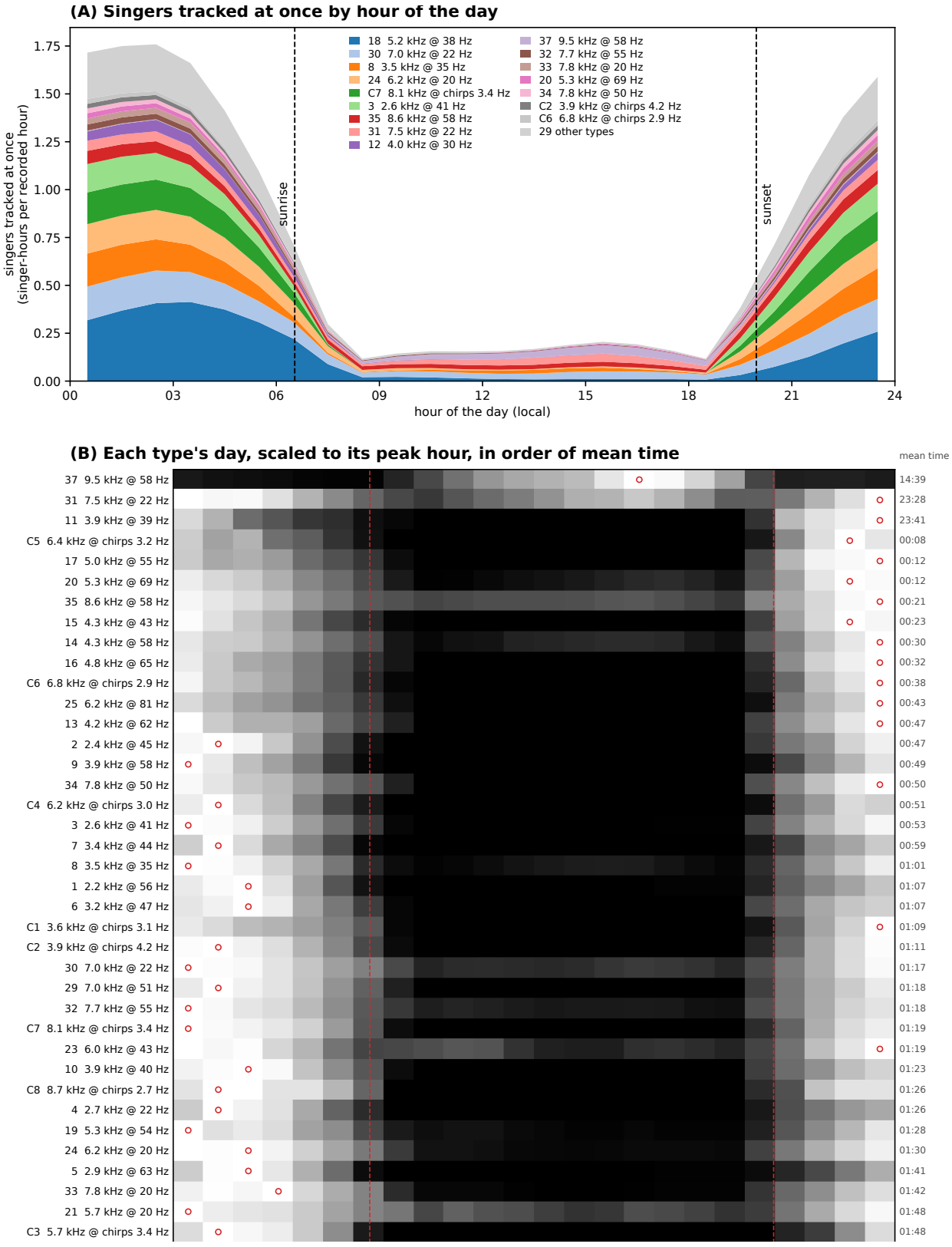

Figure 16. The types across the 24 hours of the day

.

(A) Top singers tracked by hour across year. (B) Hourly patterns of dominant types with time of peak activity shown.

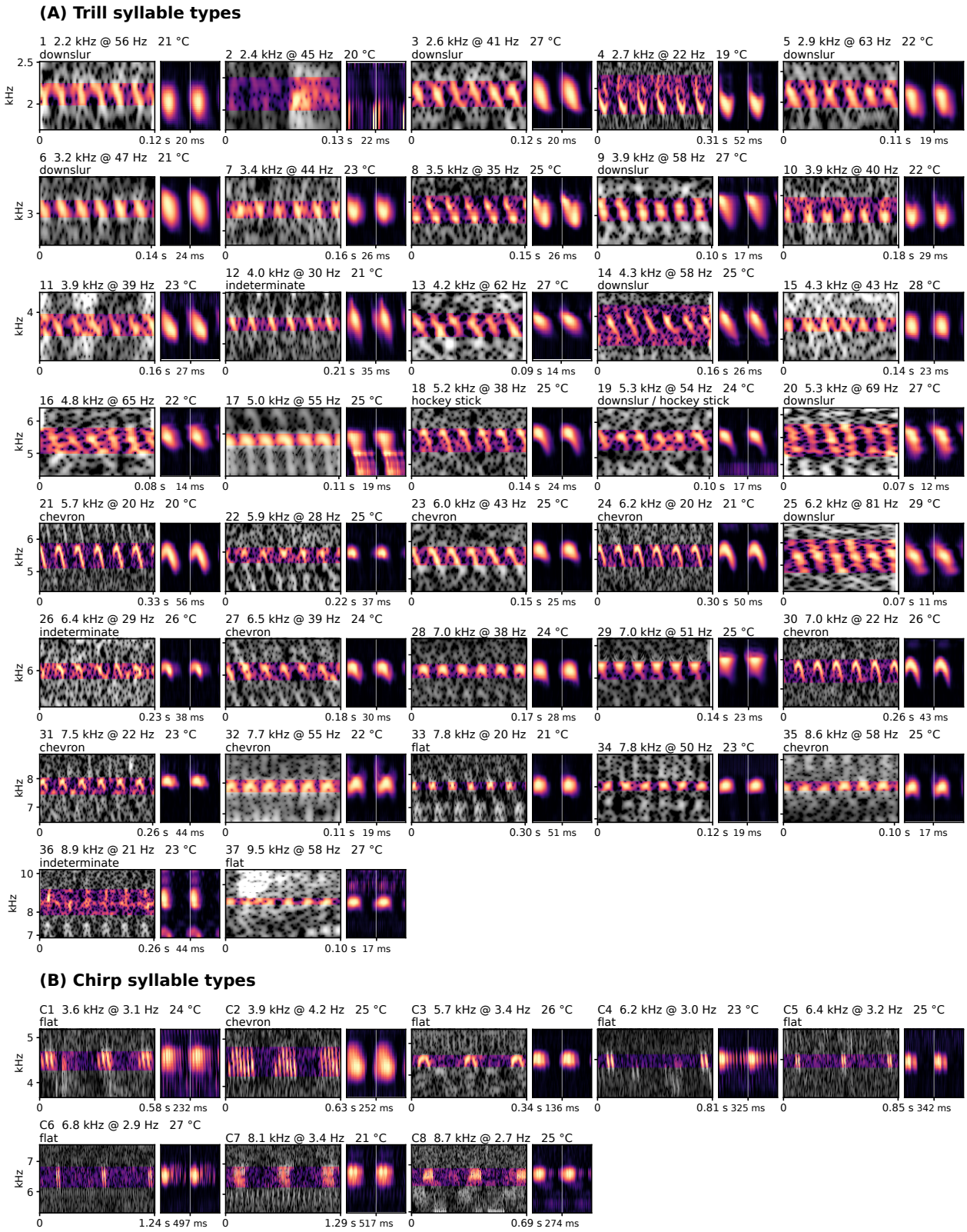

Figure 17. Syllable types across song types

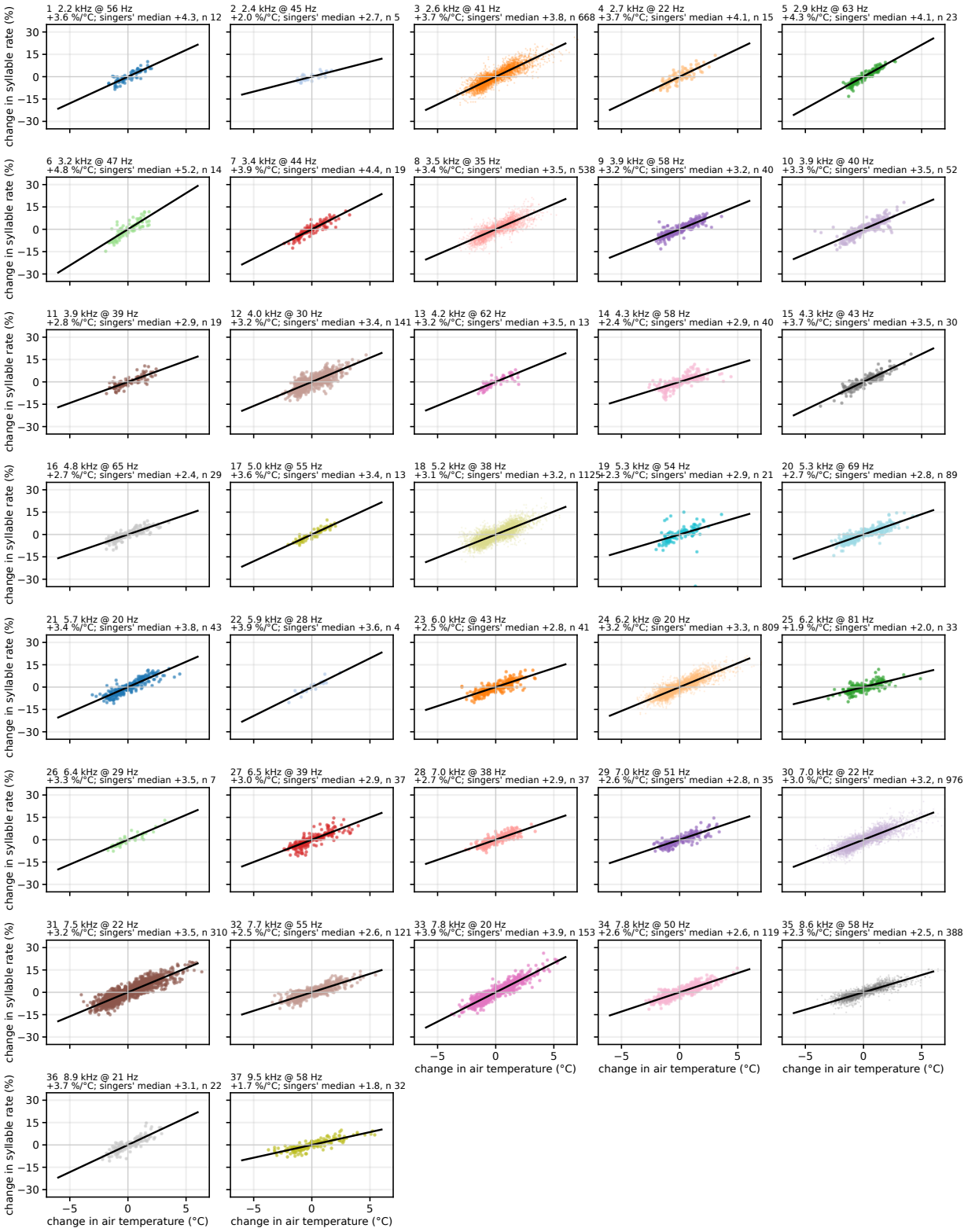

Figure 18. Syllable rate against air temperature within singers, type by type

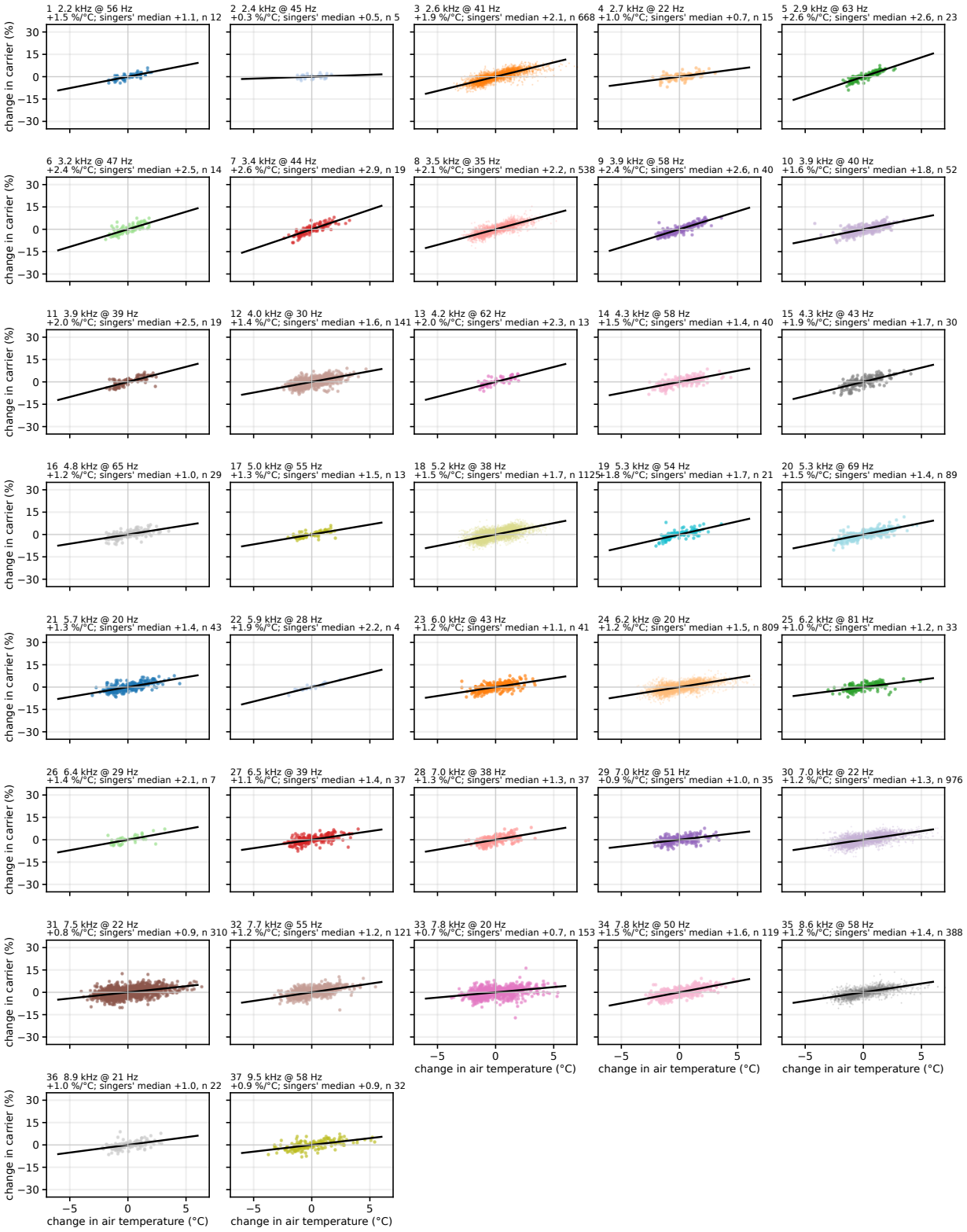

Figure 19. Carrier frequency against air temperature within singers, type by type